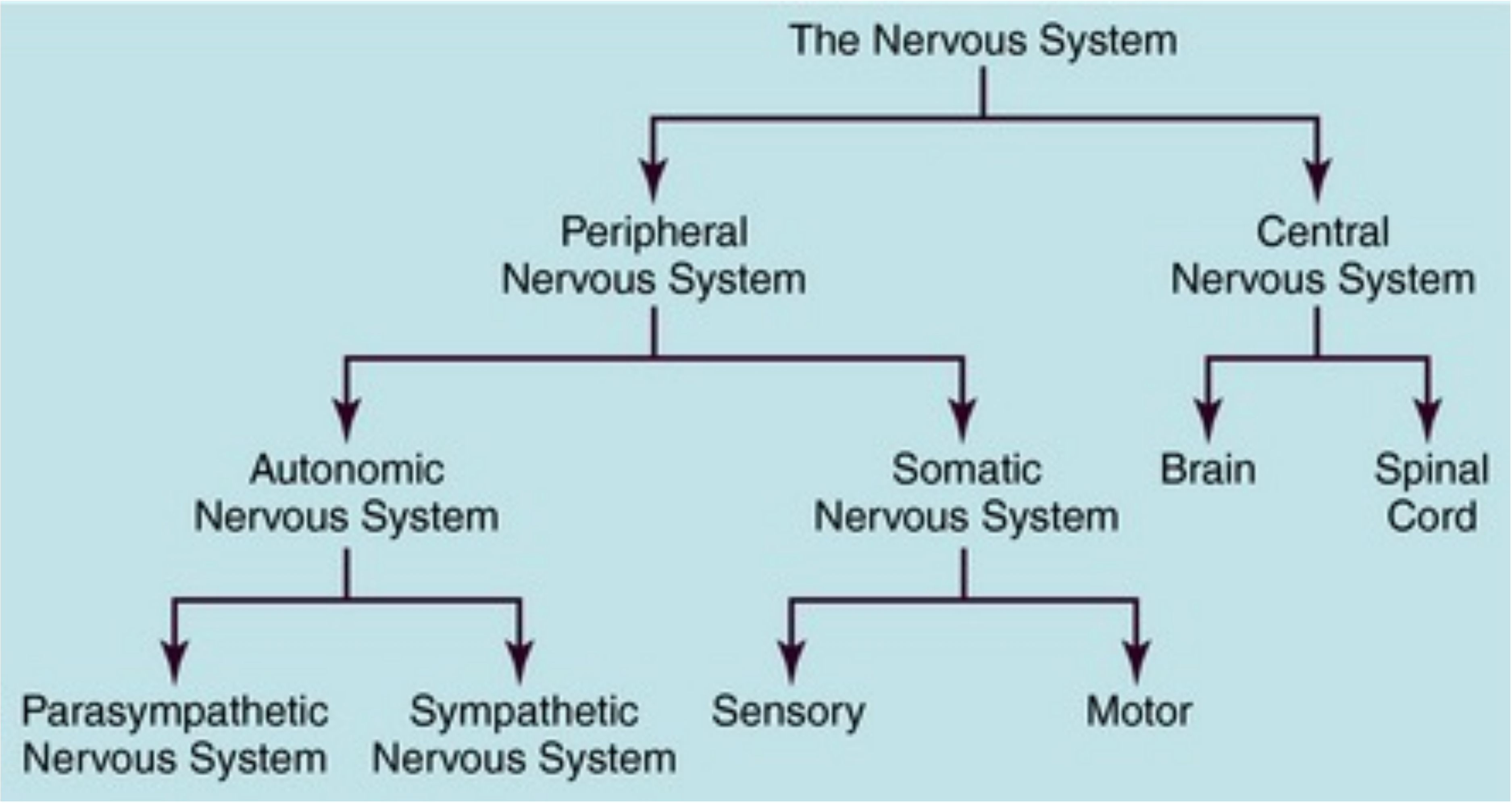
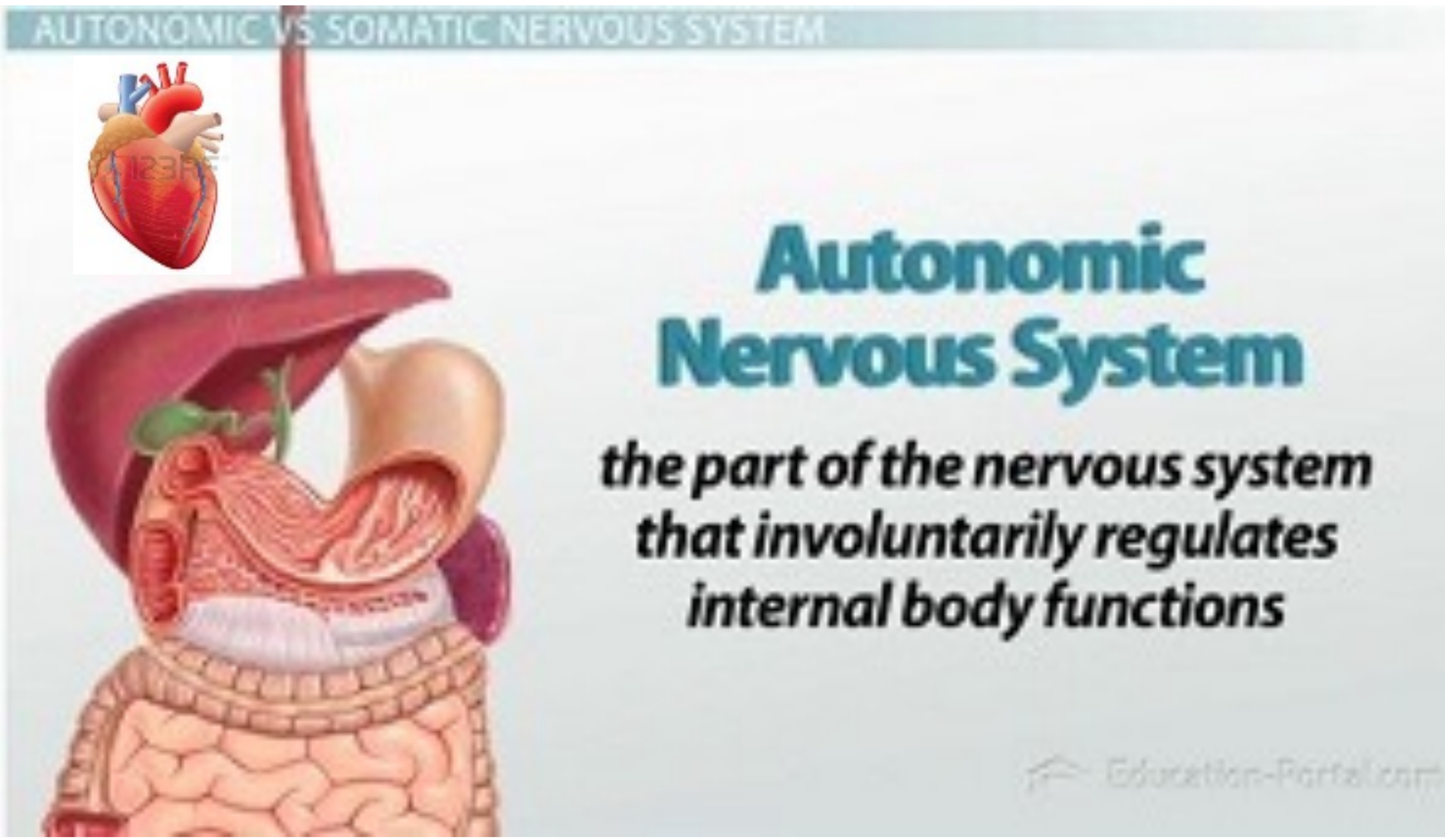
Autonomic Nervous System

Division of the nervous system




AUTONOMIC VS SOMATIC NERVOUS SYSTEM
Autonomic
Nervous System
the part of the nervous system
that involuntarily regulates
internal body functions

Afferents and Efferent

It is characterized by a two-neuron

efferent pathway:

Preganglionic neuron: Cell body

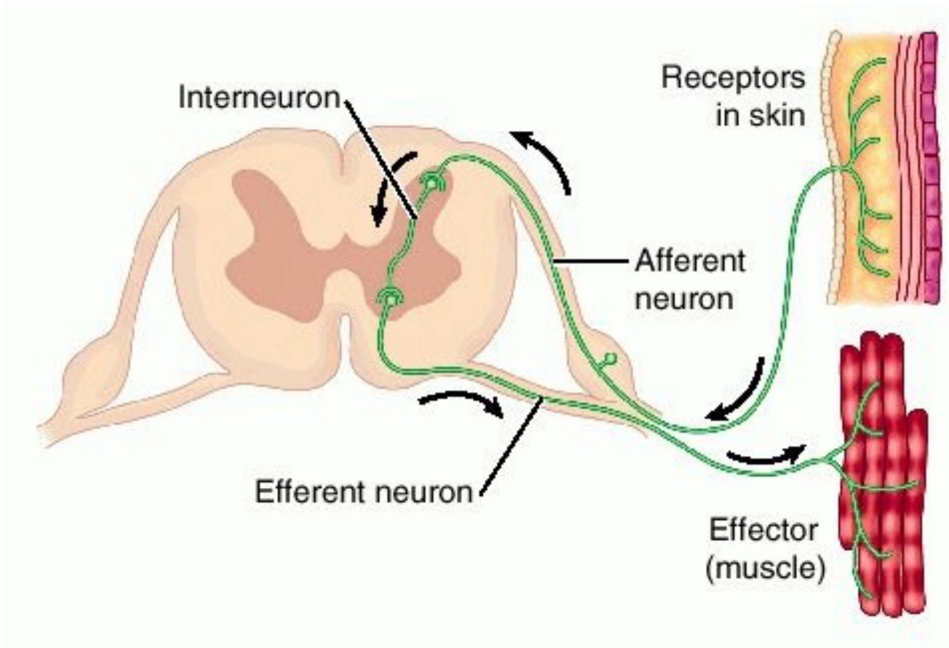
located in the CNS

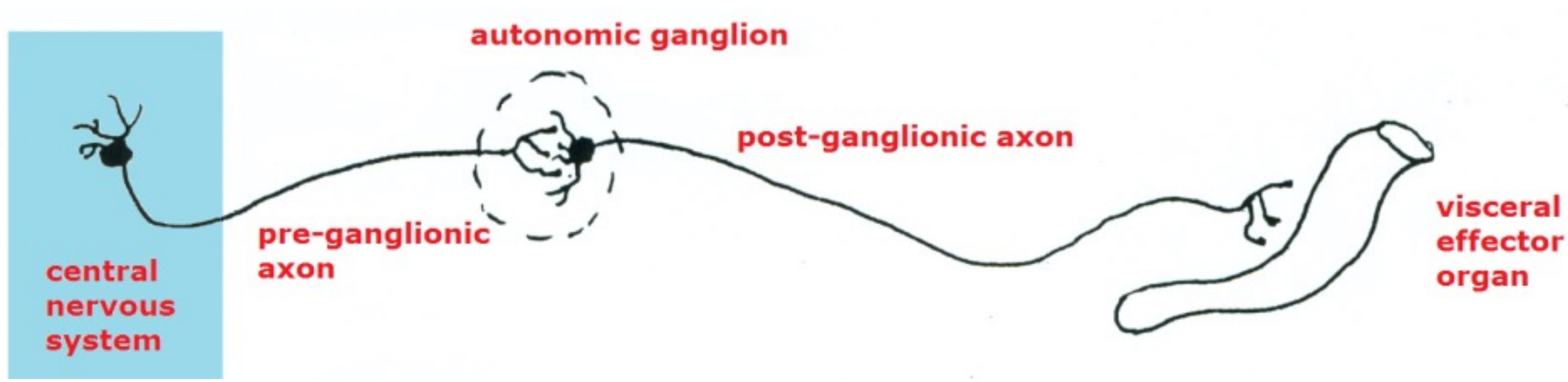
•Axon exits the CNS

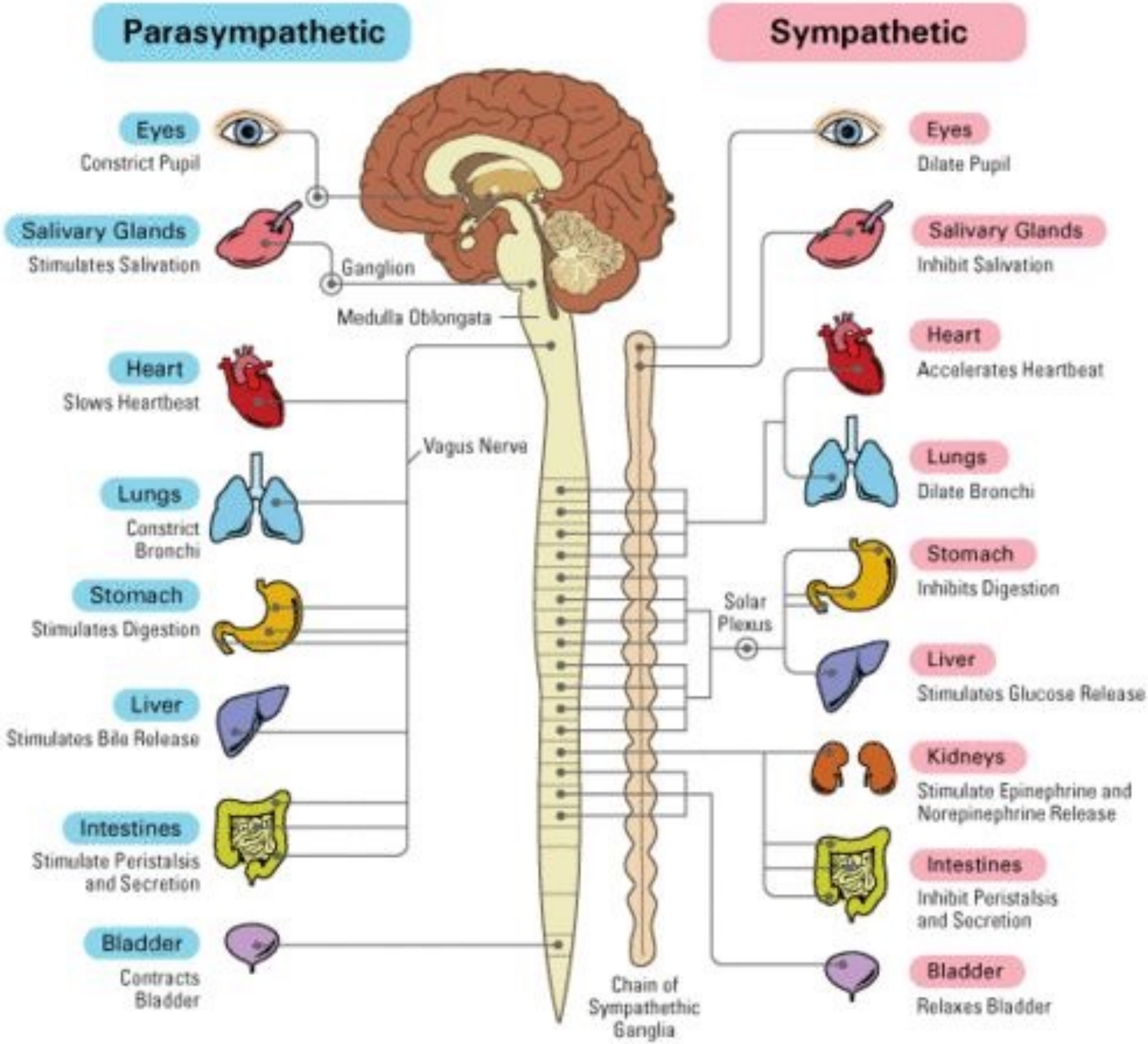
Postganglionic neuron: Cell body

located in an autonomic ganglion

•Axon supplies the target organ





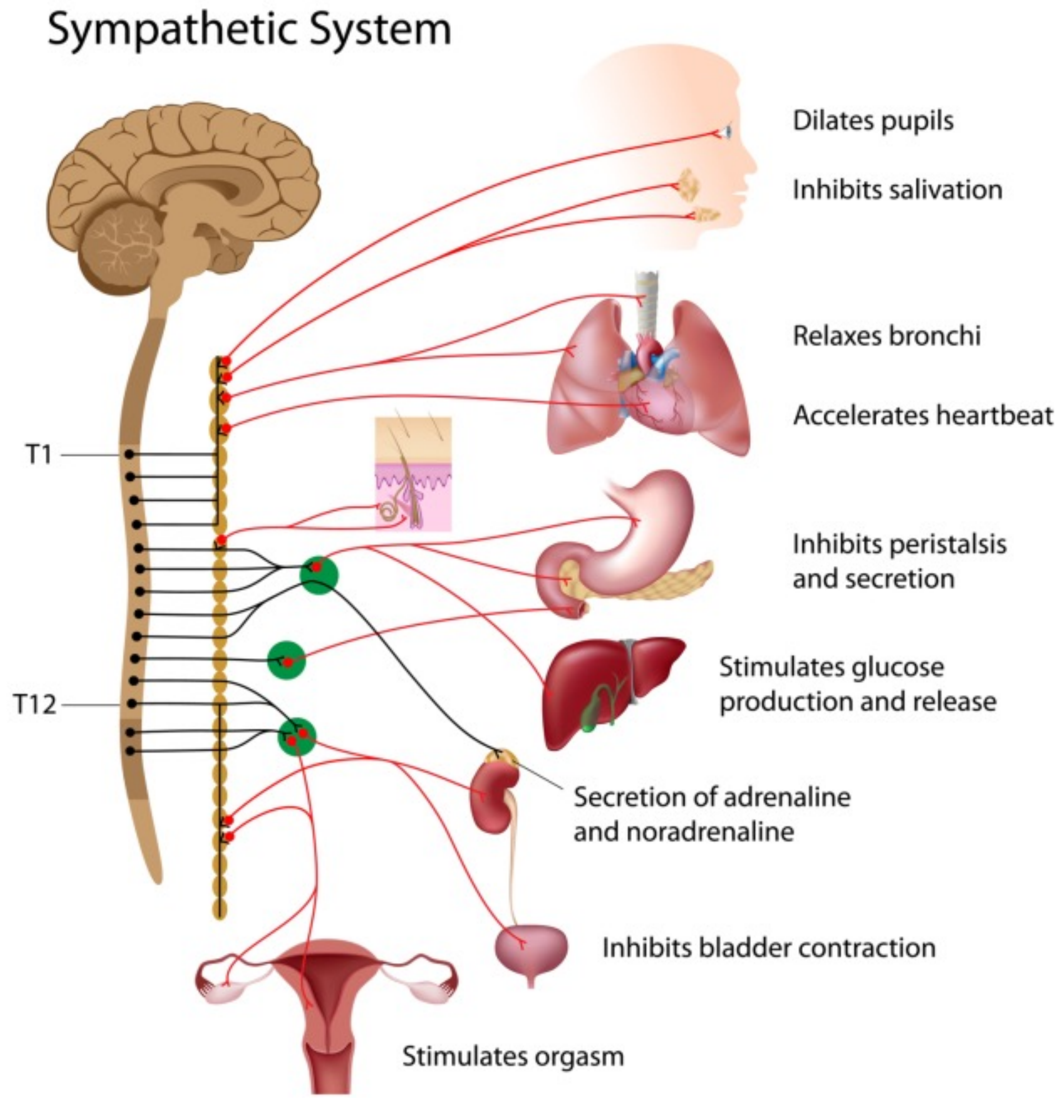


Parasympathetic
Sympathetic
Eyes
Constrict Pupil
Salivary Glands
Stimulates Salivation
Ganglion
Medulla Oblongata
Heart
Slows Heartbeat
Vagus Nerve
Lungs
Constrict Bronchi
Stomach
Stimulates Digestion
Liver
Stimulates Bile Release
Intestines
Stimulate Peristalsis and Secretion
Bladder
Contracts Bladder
Chain of Sympathethic Ganglia
Solar Plexus
Eyes
Dilate Pupil
Salivary Glands
Inhibit Salivation
Heart
Accelerates Heartbeat
Lungs
Dilate Bronchi
Stomach
Inhibits Digestion
Liver
Stimulates Glucose Release
Kidneys
Stimulate Epinephrine and Norepinephrine Release
Intestines
Inhibit Peristalsis and Secretion
Bladder
Relaxes Bladder

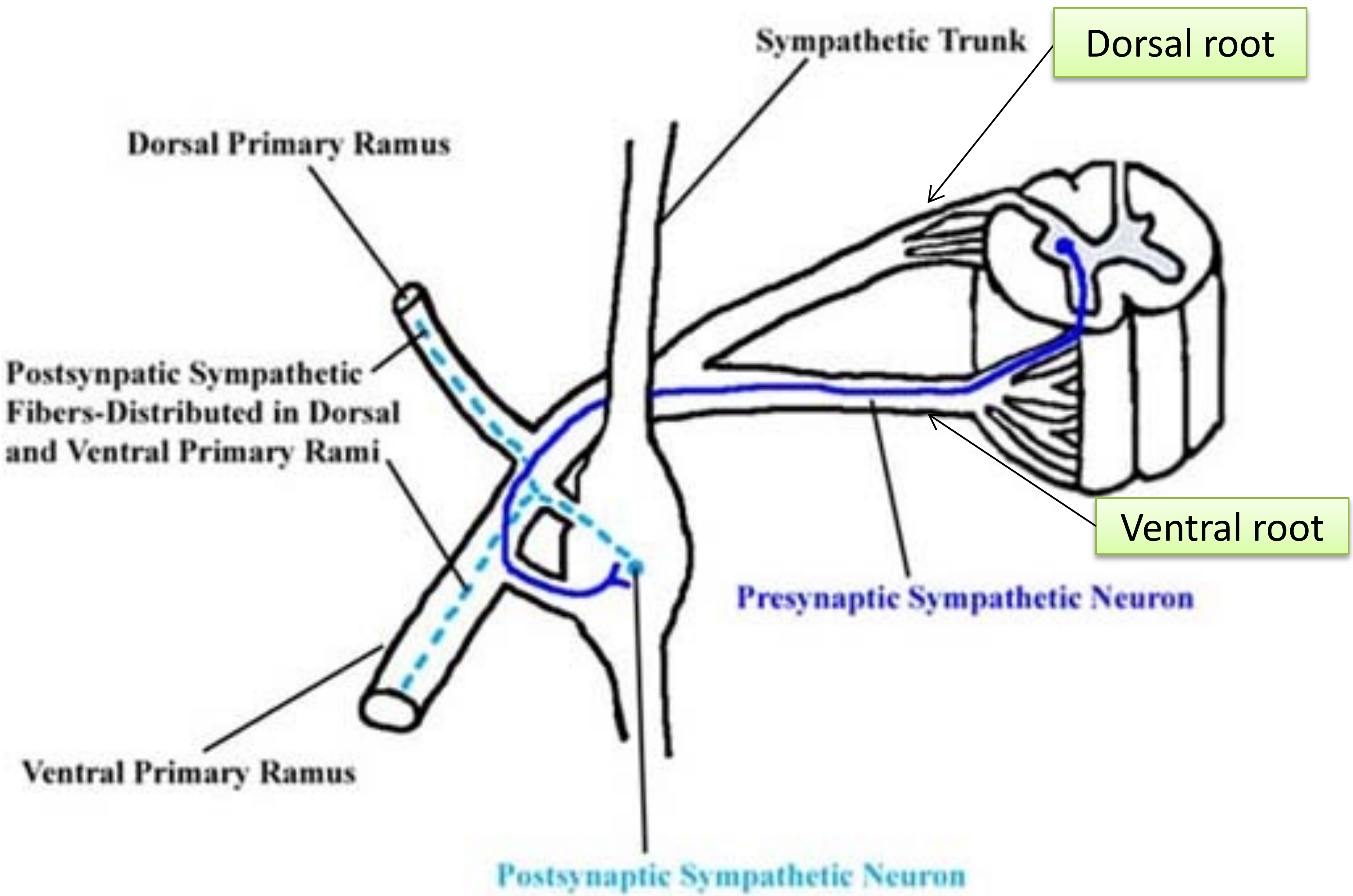
Origin Thoracolumbar Outflow

Preganglionic neurons arise from: T1–L2 spinal cord segments

Located in the lateral horn (intermediolateral cell column

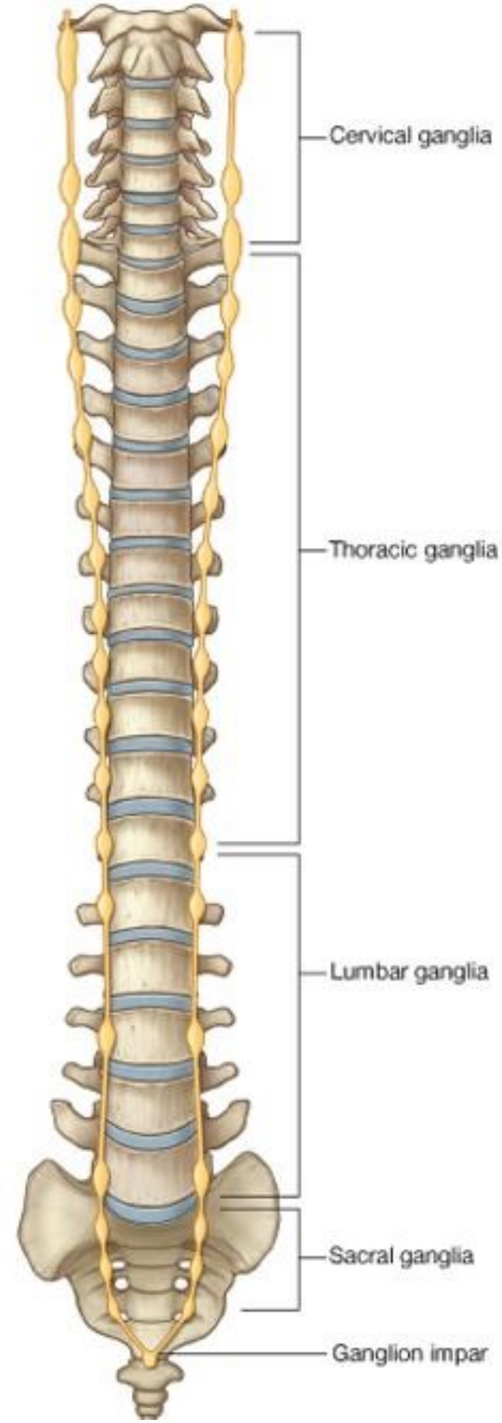


White and gray ramus



Sympathetic trunk

- Cervical part
- Thoracic part
- Lumbar part
- Sacral part



Autonomic Nervous System of Head & Neck

Sympathetic Part: Cervical Sympathetic Trunk

Cervical part of sympathetic trunk extends upward to base of skull and below to neck of 1st rib, where it becomes continuous with thoracic sympathetic trunk.

It lies directly behind internal & common carotid arteries & is embedded in deep fascia between carotid sheath & prevertebral deep fascia.

It has 3 ganglia:

1.**Superior Cervical Ganglion:** lies immediately below skull

Branches:

Internal carotid nerve : (internal carotid artery- Plexus)

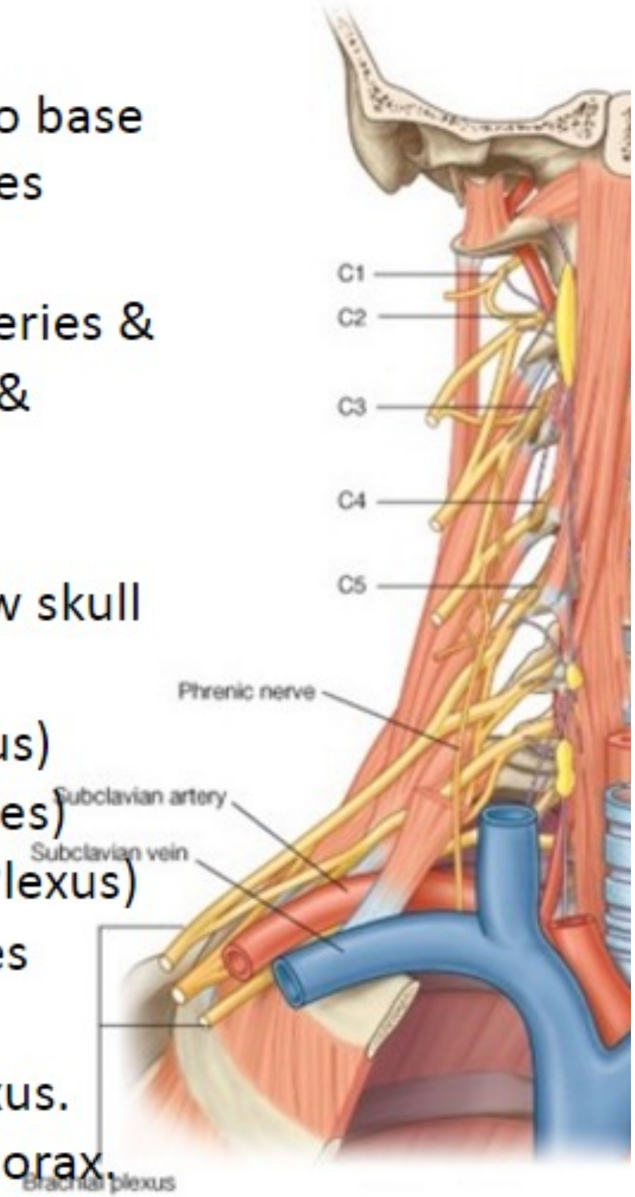
Gray rami: (upper four anterior rami of cervical nerves)

Arterial branches: (common & external carotid aa. Plexus)

Cranial nerve branches: 9th, 10th & 12th cranial nerves

Pharyngeal branches: join pharyngeal branches of glossopharyngeal & vagus n. & form pharyngeal plexus.

Superior cardiac branch: ends in cardiac plexus in thorax.



2. Middle Cervical Ganglion: lies at level of cricoid

Branches:

Gray rami: (ant. rami of 5^{th} & 6^{th} cervical nerves)

Thyroid branches : along inferior thyroid artery.

Middle cardiac branch: ends in cardiac plexus.

3. Inferior Cervical Ganglion: fused with 1^{st} thoracic ganglion to form **stellate ganglion** . It lies in interval between transverse process of C7 & neck of 1^{st} rib.

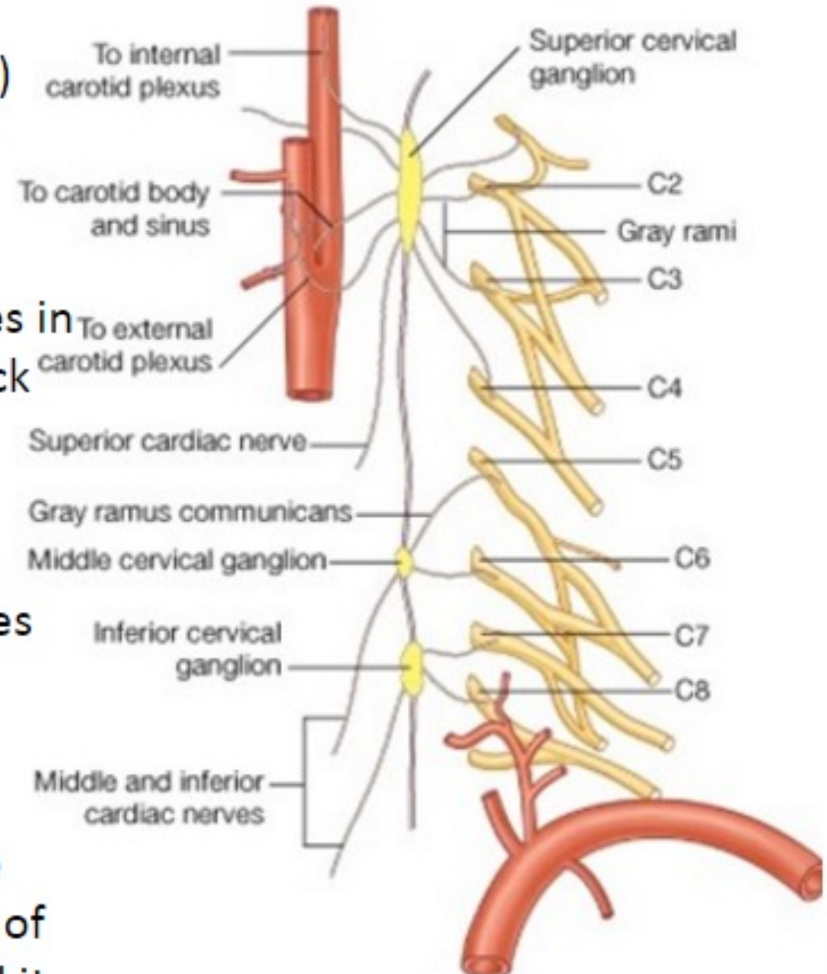
Branches:

Gray rami: (ant. rami of 7^{th} & 8^{th} cervical nerves

Arterial branches: subclavian & vertebral arteries

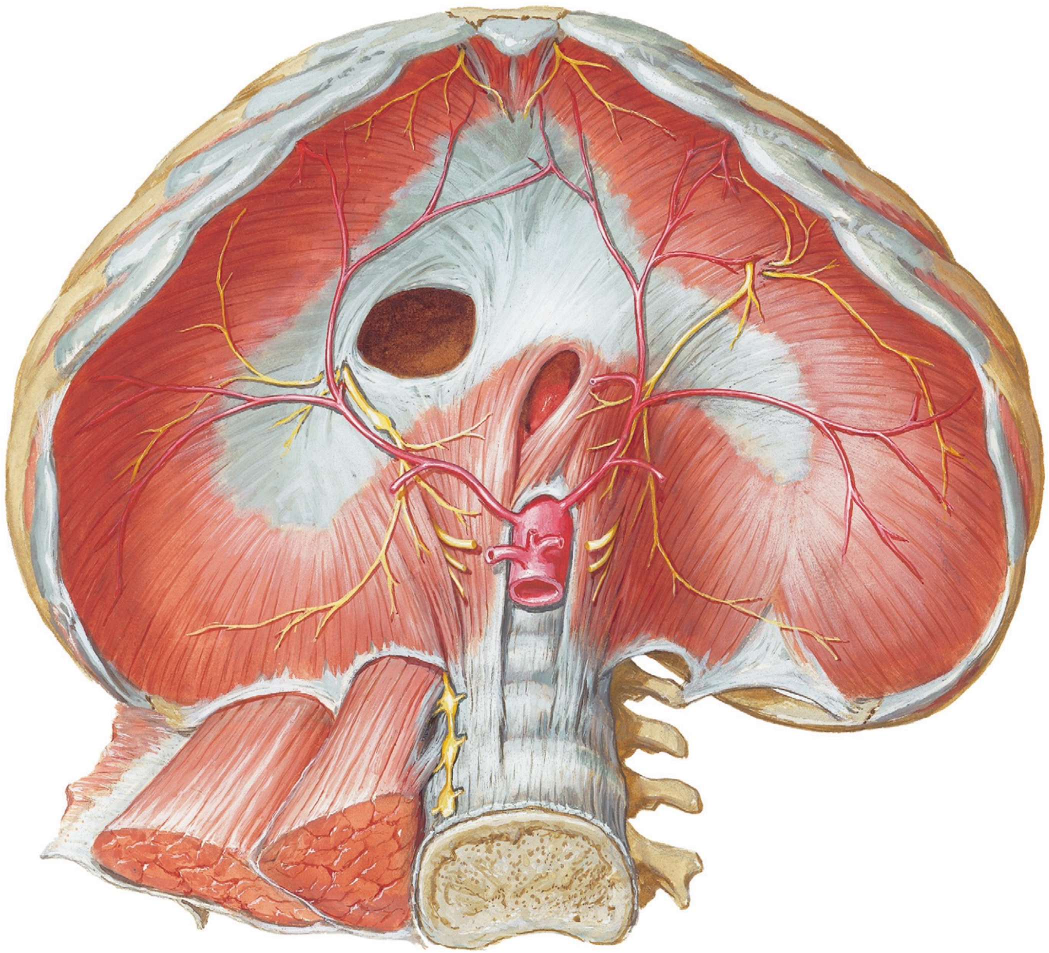
Inferior cardiac branch: join cardiac plexus.

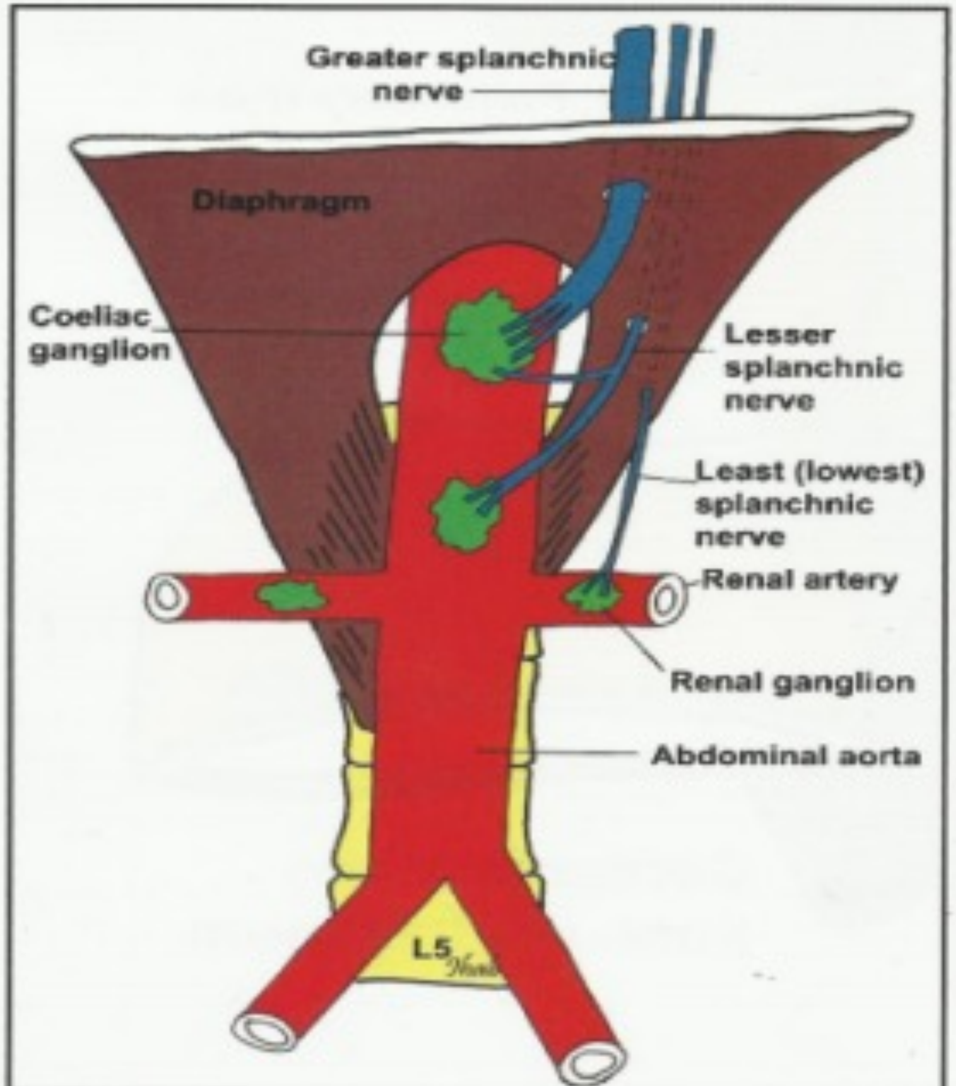
Sympathetic trunk connecting middle cervical ganglion to inferior or stellate ganglion is represented by two or more nerve bundles. The most anterior bundle crosses in front of 1^{st} part of subclavian artery and then turns upward behind it. This anterior bundle is called **ansa subclavia** .

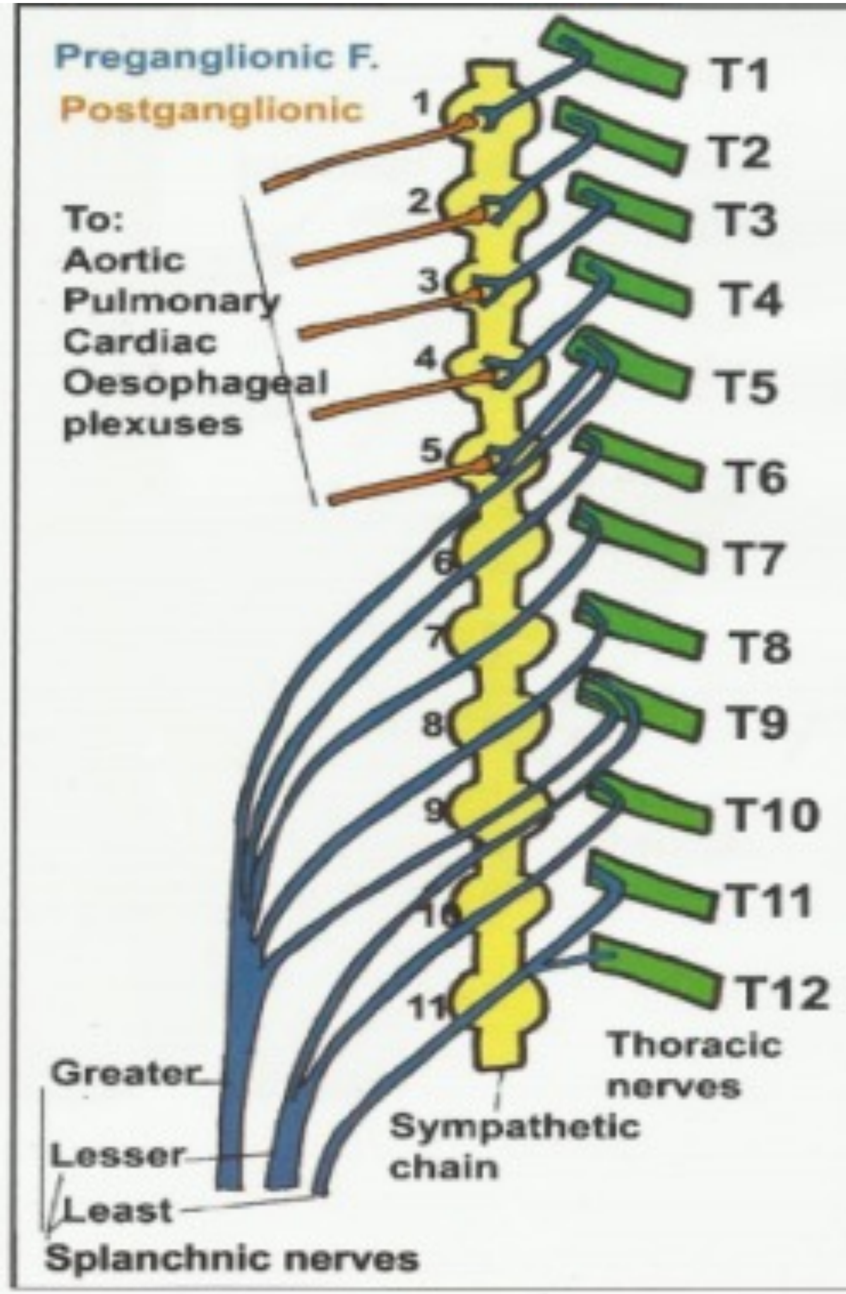


Thoracic Part of the Sympathetic Trunk

1. continuous above with the cervical and below with the lumbar parts of the sympathetic trunk.
2. It is the most laterally placed structure in the mediastinum.
3. It runs downward on the heads of the ribs.
4. It leaves the thorax on the side of the body of the T12 by passing behind the medial arcuate ligament.




Greater splanchnic nerve
Diaphragm
Coeliac ganglion
Lesser splanchnic nerve
Least (lowest) splanchnic nerve
Renal artery
Renal ganglion
Abdominal aorta
L5

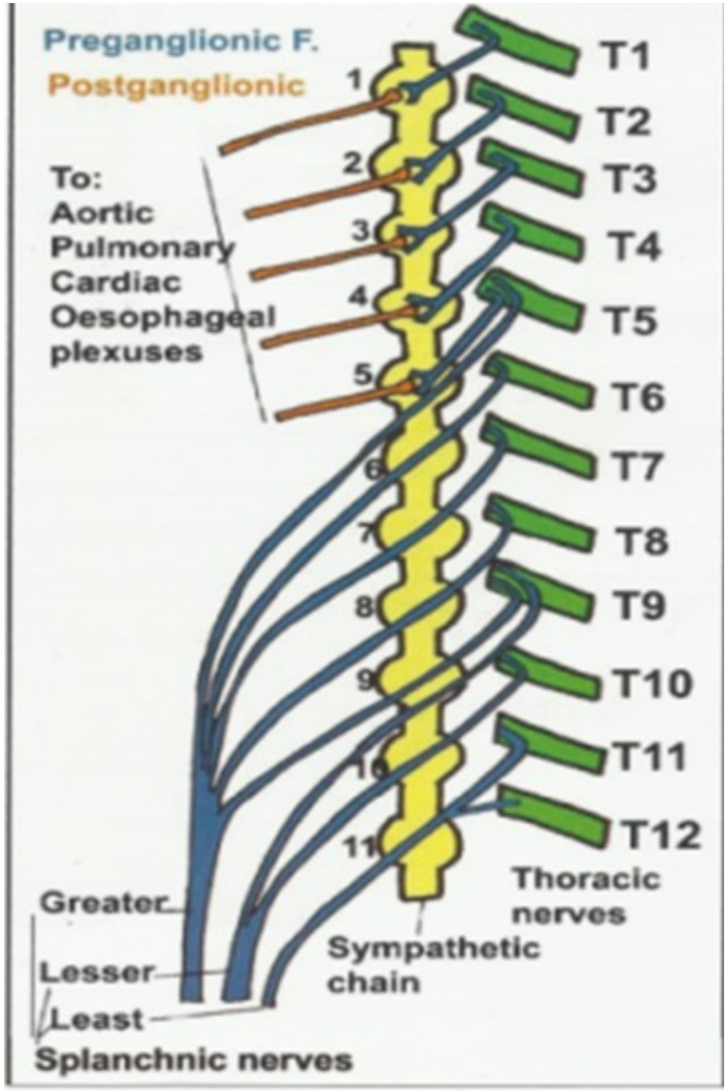

Preganglionic F.
Postganglionic
To:
Aortic
Pulmonary
Cardiac
Oesophageal
plexuses
T1
T2
T3
T4
T5
T6
T7
T8
T9
T10
T11
T12
1
2
3
4
5
6
7
8
9
10
11
Thoracic nerves
Sympathetic chain
Greater
Lesser
Least
Splanchnic nerves

Branches of thoracic part

1- Grey rami communication
to all thoracic spinal nerves

2- White rami communication

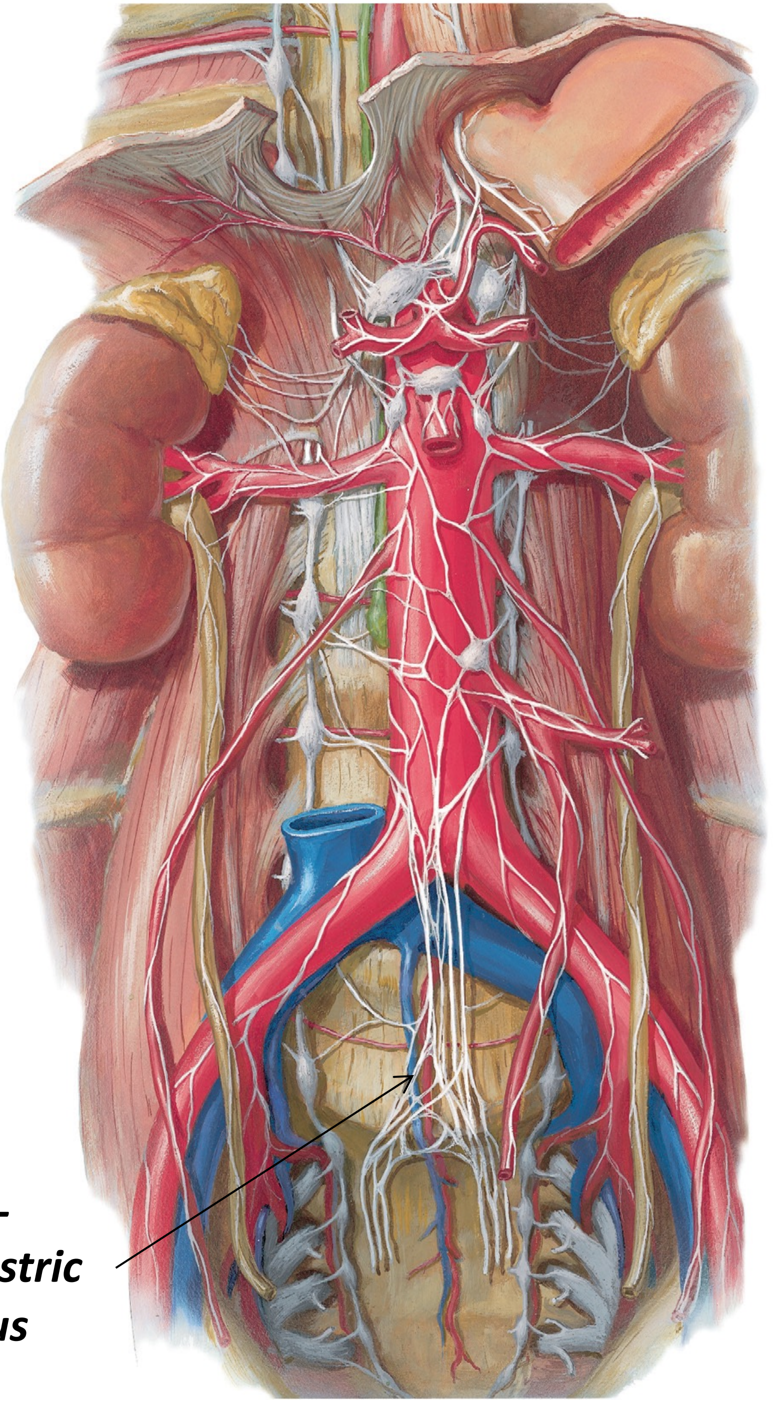
- From 1st to 5th to the (heart, lung, aorta & esophagus)
- From 5th to 9th form the greater splanchnic nerve
- From 9th to 10th form lesser splanchnic nerve
- Ganglion 11 forms the lowest splanchnic nerve



Abdominal part

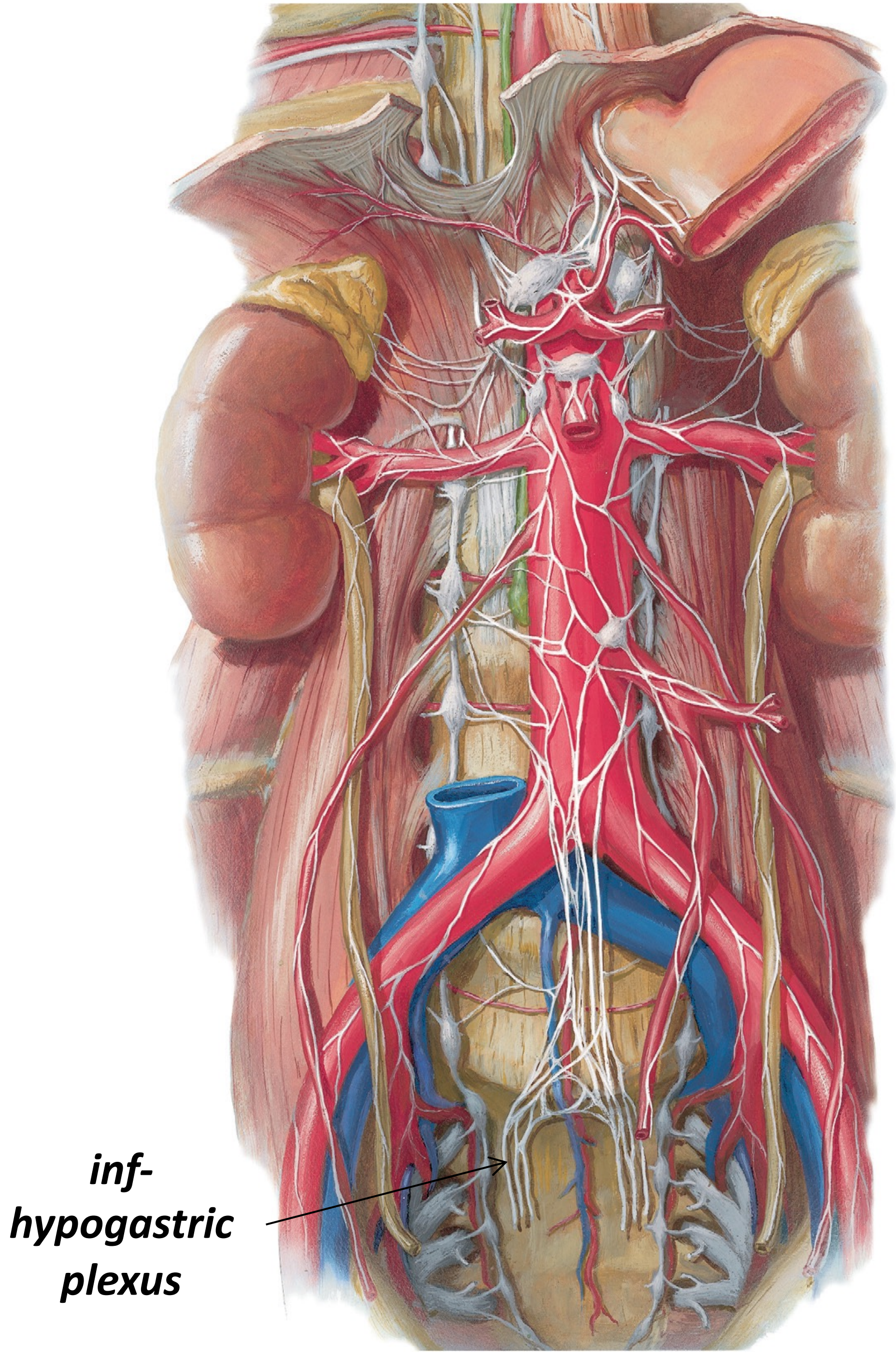
- It has 4 or five ganglia
- Only upper two ganglia receive ***white ramus communication***
- It gives

1. **Grey rami communication** to all lumbar spinal nerves
2. **Post ganglionic fibers** to aortic plexus
3. **Post ganglionic fibers** to hypogastric plexus

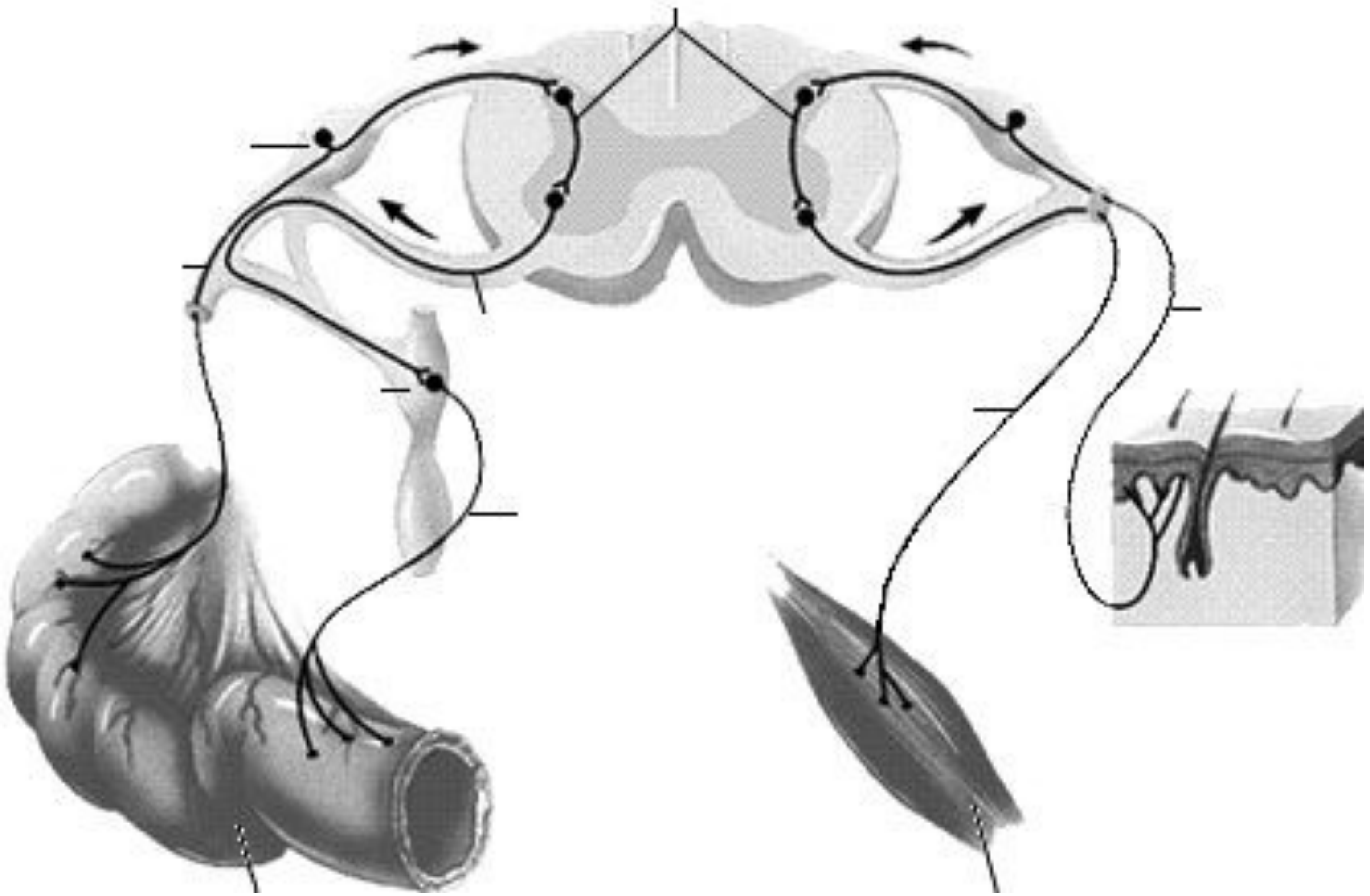


Pelvic part

- ❖ It has 4 ganglia
- ❖ It gives
- Grey rami communication to all sacral and coccygeal spinal nerves
- Post ganglionic fibers to inf- hypogastric plexus

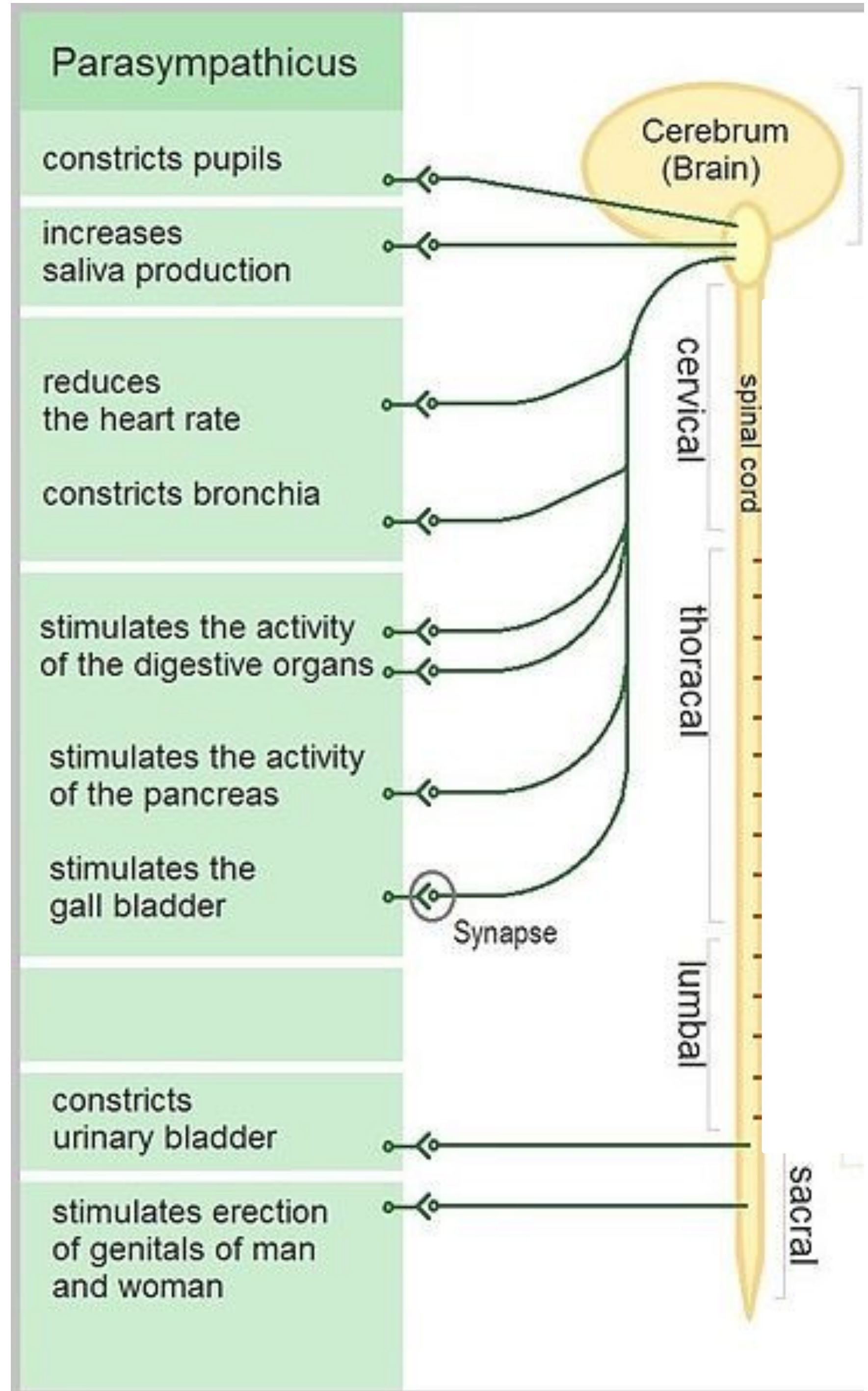


Afferent fibers



Parasympathetic sys

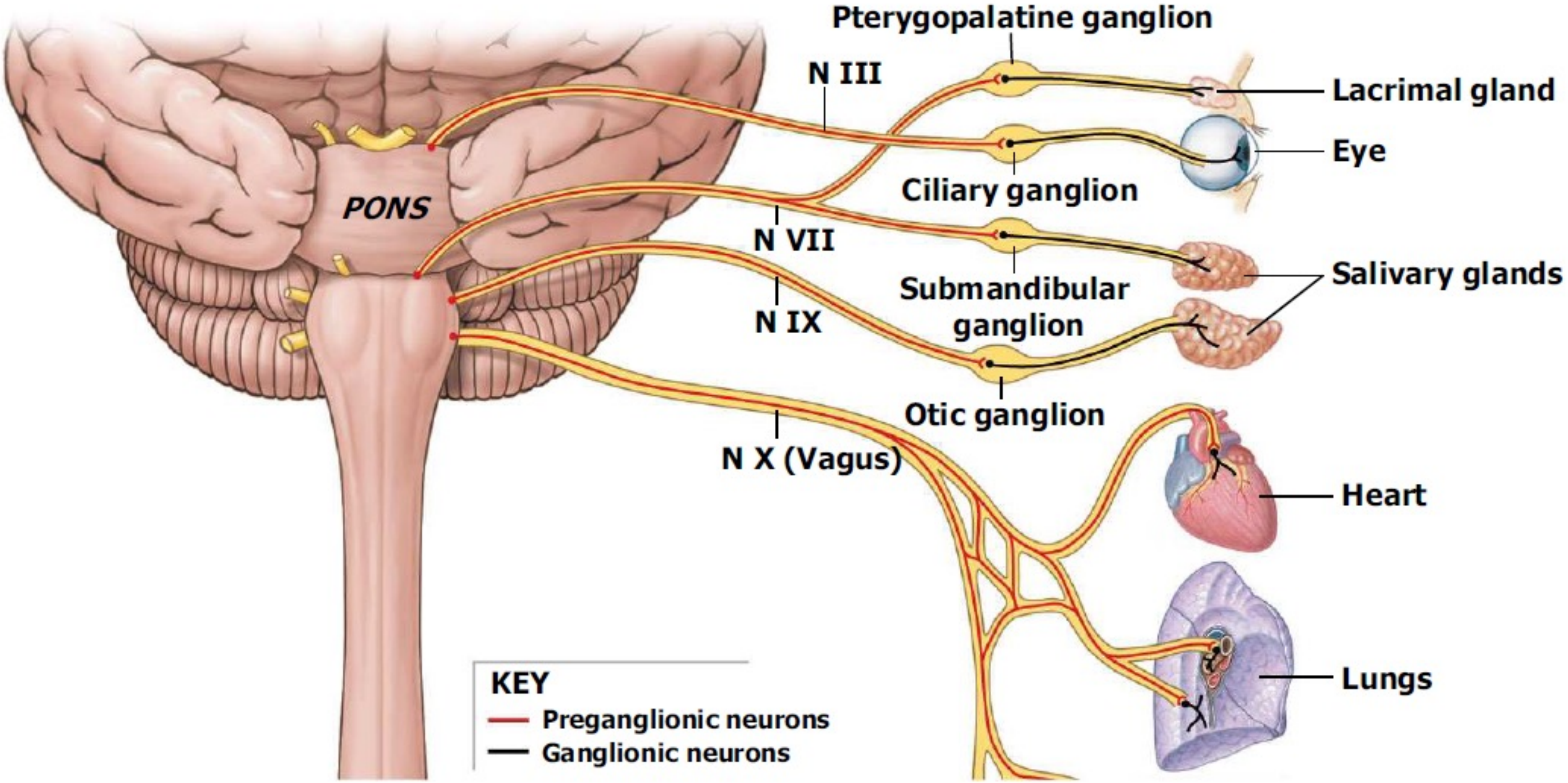
- Its action is conserving and restoring energy
- Nerve cells are located in the brain and sacral segments
- Its called the Craniosacral outflow



Parasympathetic continued

- Cranial outflow
  - III - pupils constrict
  - VII - tears, nasal mucus, saliva
  - IX – parotid salivary gland
  - X (Vagus n) – visceral organs of thorax & abdomen:
    - Stimulates digestive glands
    - Increases motility of smooth muscle of digestive tract
    - Decreases heart rate
    - Causes bronchial constriction
- Sacral outflow (S2-4): form pelvic splanchnic nerves
  - Supply 2nd half of large intestine
  - Supply all the pelvic (genitourinary) organs

- **Cranial outflow**



Thanks for your Attention